KAIZEN MATHS

Lesson starter

Algebraic substitution

Combining expressions

4 questions · about 8 minutes

Name _____

Substitute, then expand and simplify.

S1 Given $a = 2b + 2$

Write in terms of b and simplify

$5a + 4b$

S2 Given $a = 3b + 5$

Write in terms of b and simplify

$2a + 5b$

S3 Given $a = 6b + 1$

Write in terms of b and simplify

$2a + 6b$

S4 Given $a = 2b - 5$

Write in terms of b and simplify

$3a + 6b$

Show your working

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S3 Given $a = 6b + 1$

Write in terms of b and simplify

$2a + 6b$

S4 Given $a = 2b - 5$

Write in terms of b and simplify

$3a + 6b$

AS-07 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Combining expressions

Name _____

Simplify each expression.

E1 Given $a = 6b + 1$

Write in terms of b and simplify

$2a + 6b$

E2 Given $a = 2b - 5$

Write in terms of b and simplify

$3a + 6b$

AS-07 · E · 2 questions

Show your working

KAIZEN MATHS

Exit ticket

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Combining expressions

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AS-07 · E · 2 questions

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Combining expressions

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AS-07 · E · 2 questions

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AS-07 · E · 2 questions

Show your working

Teacher answers & working

AS-07 · Combining expressions

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given a=2b+2

$5a + 4b$

Write in terms of b and simplify

Substitute a=2b+2.

$5(2b + 2) + 4b$

$10b + 10 + 4b$

$(10 + 4)b + 10 = 14b + 10$

S2

Given a=3b+5

$2a + 5b$

Write in terms of b and simplify

Substitute a=3b+5.

$2(3b + 5) + 5b$

$6b + 10 + 5b$

$(6 + 5)b + 10 = 11b + 10$

S3 / E1

Given a=6b+1

$2a + 6b$

Write in terms of b and simplify

Substitute a=6b+1.

$2(6b + 1) + 6b$

$12b + 2 + 6b$

$(12 + 6)b + 2 = 18b + 2$

S4 / E2

Given a=2b-5

$3a + 6b$

Write in terms of b and simplify

Substitute a=2b-5.

$3(2b - 5) + 6b$

$6b - 15 + 6b$

$(6 + 6)b - 15 = 12b - 15$